```
 GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

volatile sig_atomic_t received = 0;

void handler(int sig) {
    received = sig;
}

int main() {
    struct sigaction sa = {0};

    sa.sa_handler = handler;
    sigemptyset(&sa.sa_mask);

    if (sigaction(SIGUSR1, &sa, NULL) == -1) {
        perror("sigaction");
        return 1;
    }

    printf("PID: %d\n", getpid());
    printf("Waiting for SIGUSR1...\n");

    while (!received)
        pause();

    printf("Received signal: %d\n", (int)received);
    return 0;
}
```

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

volatile sig_atomic_t interrupted = 0;

void handler(int sig) {
    interrupted = 1;
}

int main() {
    struct sigaction sa = {0};

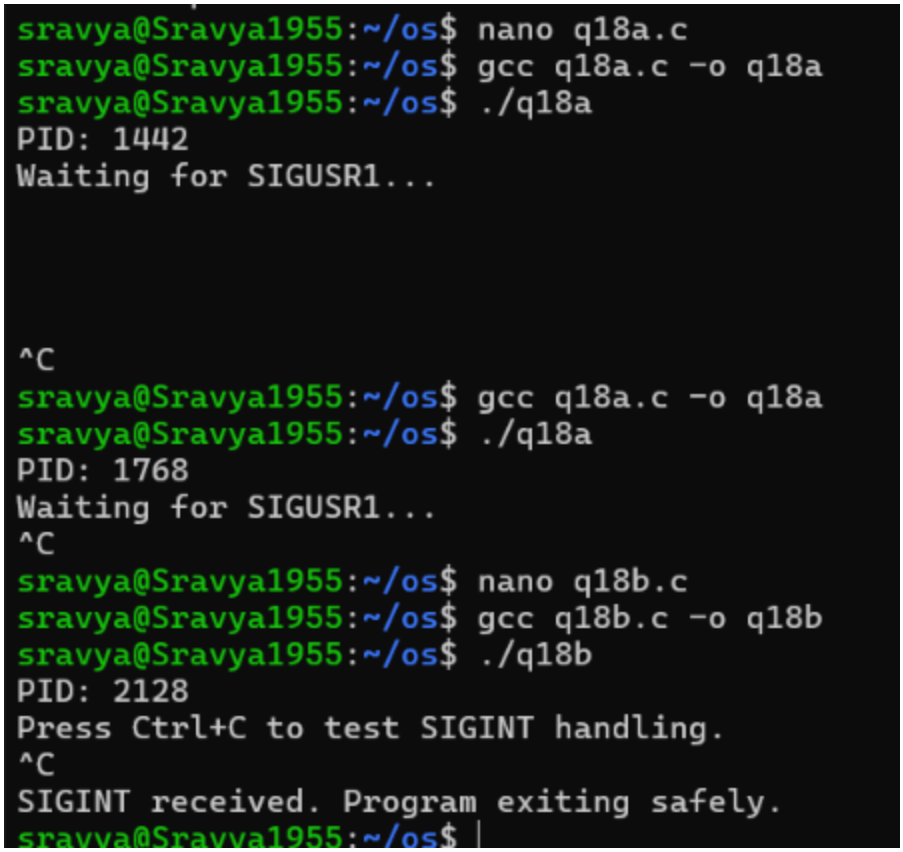
    sa.sa_handler = handler;
    sigemptyset(&sa.sa_mask);

    if (sigaction(SIGINT, &sa, NULL) == -1) {
        perror("sigaction");
        return 1;
    }

    printf("PID: %d\n", getpid());
    printf("Press Ctrl+C to test SIGINT handling.\n");

    while (!interrupted)
        pause();

    printf("\nSIGINT received. Program exiting safely.\n");
    return 0;
}
```


sravya@Sravya1955:~/os$ nano q18a.c
sravya@Sravya1955:~/os$ gcc q18a.c -o q18a
sravya@Sravya1955:~/os$ ./q18a
PID: 1442
Waiting for SIGUSR1...
^C
sravya@Sravya1955:~/os$ gcc q18a.c -o q18a
sravya@Sravya1955:~/os$ ./q18a
PID: 1768
Waiting for SIGUSR1...
^C
sravya@Sravya1955:~/os$ nano q18b.c
sravya@Sravya1955:~/os$ gcc q18b.c -o q18b
sravya@Sravya1955:~/os$ ./q18b
PID: 2128
Press Ctrl+C to test SIGINT handling.
^C
SIGINT received. Program exiting safely.
sravya@Sravya1955:~/os$